

Mini-AV: Lightweight Python Antivirus Scanner

An overview of the tool's architecture and features



Lightweight AV Engine

Designed for fast scanning and recursive archive handling using Python and 7-Zip CLI.



Magic-Header Based Detection

Identifies archives based on byte signatures instead of unreliable file extensions.



Malicious Hash Matching

Scans and compares multiple cryptographic hashes against known malware fingerprints.

Magic-Header Detection of Archives

Bypassing misleading file extensions

- **Signature-Based Identification:** Reads initial byte patterns (magic headers) to recognize archive types like ZIP, RAR, 7z, TAR, GZIP.
- **Offset-Aware Logic:** Handles special cases like TAR, where signature resides at byte offset 257.
- **Avoids Extension Reliance:** Improves reliability by not depending on potentially incorrect or spoofed file extensions.



Photo by Mika Baumeister on Unsplash

7-Zip Powered Decompression

Unified CLI-based archive handling

- **7-Zip Integration:** Leverages the `7z` CLI tool to extract files of multiple formats including ZIP, RAR, GZIP, BZIP2, XZ, and 7z.
- **Consistent Extraction Logic:** Simplifies decompression workflow across formats using a single command: `7z x -o`
- **No Python Dependency on Format Modules:** Avoids format-specific Python libraries by relying entirely on an external, format-agnostic decompression engine.

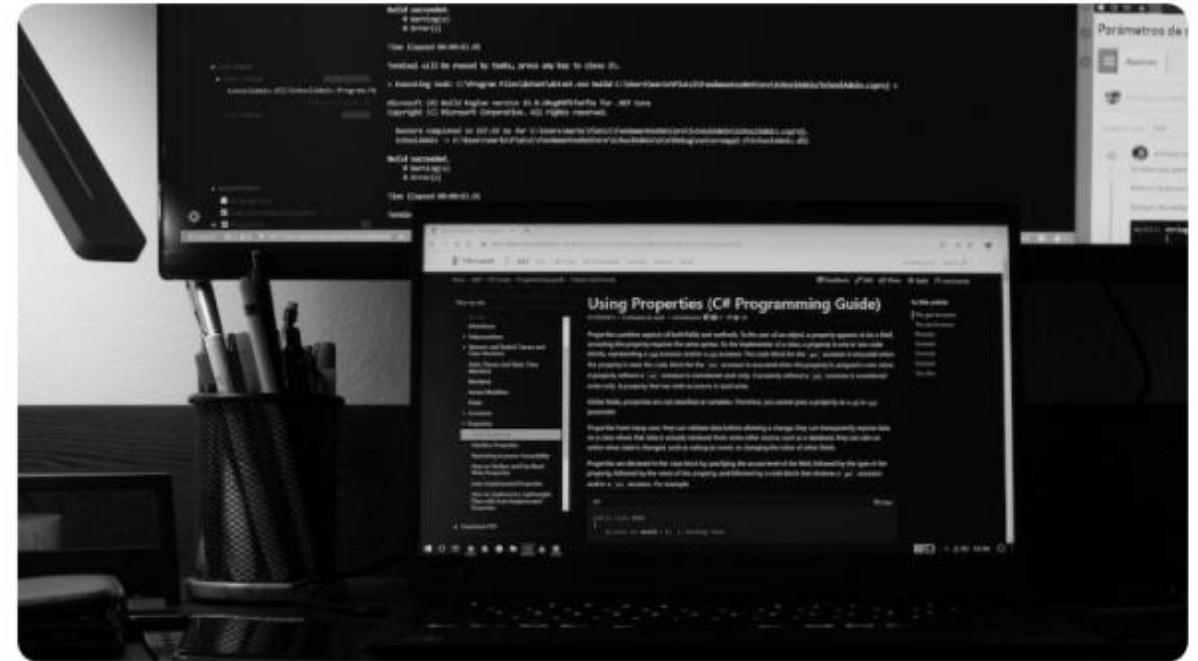


Photo by Mario Menjívar on Unsplash

Malicious File Detection via Hash Comparison

Flagging known threats with precision

- **Multi-Hash Computation:** Generates MD5, SHA1-SHA512, SHA3 family, and CRC32 hashes for all scanned files.
- **Malware Fingerprint Matching:** Compares computed hashes against a curated list of known malicious values.
- **Colorized Alerts with Colorama:** Uses terminal colors to clearly highlight filename, algorithm, and matched hash.



Photo by Hosna Asgari on Unsplash

Live Demo: Mini-AV in Action

Terminal-based scanning of nested archives



User Prompt & Path Input

Script starts by prompting for the target file or folder path to scan.



Real-Time Extraction & Analysis

Displays decompression and hashing results in the terminal with highlights.



Malware Hit Feedback

Instant color-coded alerts when malicious hashes are detected.

```
PS C:\Users\1337Muhammad> \Mini-AV> python .\mini-av-1.py
Enter the full path to scan: \Mini-AV\Task2.zip
Task2/5869342.exe
crc 8ace3486

Task2/8572213.ini
sha512 69d55495b7d59d72d32a07755a39197617927248edc3b72fb476f3ff3d05bd33c967928acf05df93897aedd58c75064694e6bc7a9b0a6aa618a5987ffadfed2c

Task2\0923131\0923131.dll
sha256 f11fa868ac3dee1e5fbd985fe15ba6d34c7ec0abb47babe0d34a35514c49c86a

Task2\4132356\4132356.exe
md5 6b2f645881bed988d32c4f7241f3a8dd

Task2\5376489\5376489.dat
sha384 9fd20defd9b97add19d682d83e1a8ed5f20496c2d03110e15a6f7ba1c2e2999f0195f974fa0de8174edf2f09dce8c9f2

Completed.
```

Planned Improvements for Mini-AV

Next steps for enhanced functionality and usability



Optional Path Cleanup

Add a `--remove-path` argument to auto-delete extracted folder structure after scanning.



VirusTotal Integration

Enable API-based scanning of file hashes for cloud-based threat intelligence.



Command-Line Flexibility

Support `hashes_db_file` and `scan_target` as CLI arguments for automation.